Coupling between geochemical reactions and multicomponent gas and solute transport in unsaturated media: A reactive transport modeling study

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[1] The two-way coupling that exists between biogeochemical reactions and vadose zone transport processes, in particular gas phase transport, determines the composition of soil gas. To explore these feedback processes quantitatively, multicomponent gas diffusion and advection are implemented into an existing reactive transport model that includes a full suite of geochemical reactions. Multicomponent gas diffusion is described on the basis of the dusty gas model, which accounts for all relevant gas diffusion mechanisms. The simulation of gas attenuation in partially saturated landfill soil covers, methane production, and oxidation in aquifers contaminated by organic compounds (e.g., an oil spill site) and pyrite oxidation in mine tailings demonstrate that both diffusive and advective gas transport can be affected by geochemical reactions. Methane oxidation in landfill covers reduces the existing upward pressure gradient, thereby decreasing the contribution of advective methane emissions to the atmosphere and enhancing the net flux of atmospheric oxygen into the soil column. At an oil spill site, methane oxidation causes a reversal in the direction of gas advection, which results in advective transport toward the zone of oxidation both from the ground surface and the deeper zone of methane production. Both diffusion and advection contribute to supply atmospheric oxygen into the subsurface, and methane emissions to the atmosphere are averted. During pyrite oxidation in mine tailings, pressure reduction in the reaction zone drives advective gas flow into the sediment column, enhancing the oxidation process. In carbonate-rich mine tailings, calcite dissolution releases carbon dioxide, which partly offsets the pressure reduction caused by O_2 consumption.

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1. Introduction

[2] The investigation and modeling of gas transport and reaction processes in the vadose zone has received increasing interest in recent years through studies of sites contaminated by volatile organic compounds [*Falta et al.*, 1989; *Mendoza and Frind*, 1990], including those with buildings affected by the intrusion of contaminated soil gas [*Garbesi et al.*, 1993; *Fischer et al.*, 1996], and sites affected by generation of acid mine drainage [*Wisotzky*, 1994; *Ritchie*, 2003, *Binning et al.*, 2007]. Additional motivation stems from the growing concern about emission of greenhouse gases to the atmosphere from sanitary landfills [*Metcalfe and Farquhar*, 1987; *Scheutz and Kjeldsen*, 2003] or leaking from reservoirs for CO_2 sequestration [*Klusman*, 2003; *Altevogt and Celia*, 2004]. The vadose zone plays an important role because it can act as a buffer zone for contaminants on their way to the water table, but can also attenuate the emission of contaminants leaving the subsurface environment through the gas phase. In many cases, these contaminants are present as part of a multicomponent mixture of gases and tend to react with atmospheric gases to produce chemical species with different chemical and physical properties. As a result, both transport processes and chemical transformations affect the composition of the gas phase, resulting in a dynamic and inherently coupled system. Examples include gas production and release in partially saturated landfills [*De Visscher et al.*, 1999; *Scheutz and Kjeldsen*, 2003], methanogenesis and methane oxidation in aquifers contaminated by organic compounds [*Amos et al.*, 2005], and the oxidation of sulfide minerals in mine waste deposits [*Ritchie*, 2003].

[3] In many cases, diffusion is considered the most relevant gas transport mechanism, and it is often the only mechanism included in models of vadose zone gas transport [*Elberling et al.*, 1994; *Mayer et al.*, 2002]. Gas diffusion as described by Fick's law affects each gas in a mixture differently because diffusion is driven by the concentration gradient of each individual gas and is therefore defined as a separative transport process [*Thorstenson and Pollock*, 1989]. However, in multicomponent mixtures diffusion of each gas component is also a function of the concentration of all other gas components and may include a nonseparative component [*Thorstenson and Pollock*, 1989]. Multicomponent diffusion in porous media is therefore better

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described by the dusty gas model (DGM) [*Thorstenson and Pollock*, 1989; *Massmann and Farrier*, 1992], although in advection-dominated systems or when diffusing gas species are dilute in the bulk gas phase, a transport model incorporating Fick's law may give similar predictions to those of the DGM [*Fen and Abriola*, 2004]. Mechanical dispersion is negligible in many gas transport applications, because its contribution to dispersion is typically small [*Massmann and Farrier*, 1992]. However, for cases involving high velocities, such as air injection/extraction systems, the inclusion of mechanical dispersion may be required.

[4] Gas advection is nonseparative in nature because it affects all gases in a mixture equally and is often a relevant gas transport process, in particular because even small pressure gradients (<1 Pa m^{-1}) can cause significant viscous fluxes [*Thorstenson and Pollock*, 1989]. Despite its potential significance, gas advection has only been included in a limited number of models. These models show that viscous gas transport can be caused by a variety of processes including the injection and extraction of air or vapor [e.g. *Massmann*, 1989; *Falta et al.*, 1992], the volatilization of organic compounds [e.g., *Falta et al.*, 1989; *Mendoza and Frind*, 1990; *Gaganis et al.*, 2004], sustained underpressurization relative to atmospheric conditions in basements of buildings and the adjacent sediments [e.g., *Hers et al.*, 2002; *Abreu and Johnson*, 2005], barometric pumping [e.g., *Massmann and Farrier*, 1992], displacement due to infiltrating recharge water [e.g., *Celia and Binning*, 1992], and temperature changes [e.g., *White*, 1995].

[5] In low-permeability media (permeabilites smaller than 10^{-13}–10^{-14} m^2), molecule-soil interactions become relevant and diffusive and advective fluxes are intrinsically coupled through Knudsen diffusion [*Thorstenson and Pollock*, 1989]. Knudsen diffusion has been considered in the characterization of the composition and movement of gases in the unsaturated zone through the use of the DGM; examples of applications include the simulation of transport and consumption of atmospheric oxygen near a lignite mine [*Thorstenson and Pollock*, 1989], the effects of temporal variations of atmospheric pressure on gas composition [*Massmann and Farrier*, 1992], and forced advection and removal of benzene and toluene in a soil venting system [*Sleep*, 1998].

[6] In addition to gas transport, interphase mass transfer processes, solute transport, and biogeochemical reactions also affect the composition of the gas phase. Mass transfer between the gaseous phase and the aqueous phase is often sufficiently fast that an equilibrium approach is valid [*White*, 1995; *Saaltink et al.*, 2004]. At equilibrium, the ratio between gaseous and aqueous concentrations is known and equal to Henry's law constant. Chemical reactions are described using aqueous species as primary variables in almost all multiphase reactive transport models [e.g., *Mayer et al.*, 2002; *Saaltink et al.*, 2004]. Mass in the aqueous phase is also exchanged with the solid phase through surface and precipitation-dissolution reactions.

[7] Biogeochemical reactions affect the composition of the gas phase by transforming reactive gas species into product species with properties such as molecular weight or aqueous solubility different than those of the reactants. In consequence, chemical transformations change the composition and properties (e.g., density, viscosity, and diffusion coefficients) of the gas mixture affecting the variables that control the transport processes such as pressure and concentrations. Therefore a two-way coupling exists between gas transport and biogeochemical reactions that affects the composition of the gas phase. For instance, aerobic oxidation of methane mediated by methanotrophic bacteria produces gaseous species (carbon dioxide) with a larger aqueous solubility than the reactants. The diffusive flux of nitrogen is also affected because the binary diffusion coefficient is reduced from 2.1×10^{-5} m^2 s^{-1} for N_2-CH_4 to 1.6×10^{-5} m^2 s^{-1} for N_2-CO_2. In addition, a decrease in gas volume caused by this reaction also contributes to a decrease in gas phase pressure and hence causes viscous gas transport into the reaction zone. Oxygen supplied by this flow may aid in sustaining the reaction and will contribute to maintaining the dynamic balance between transport and reaction processes.

[8] Models that consider coupled gas transport and reactions have been developed previously, but are designed for the solution of specific problems. For example, models of coupled methane oxidation and transport in landfill covers include only the gaseous phase and were designed to deal exclusively with methane oxidation [*Hilger et al.*, 1999; *Stein et al.*, 2001; *De Visscher and Van Cleemput*, 2003]. Recently, the role of sulfide mineral oxidation on gas transport of O_2 and N_2 has been investigated by *Binning et al.* [2007].

[9] Multiphase flow and transport models have been developed mostly for simulating the fate and attenuation of volatile organic contaminants in the vadose zone [e.g., *Falta et al.*, 1989; *Mendoza and Frind*, 1990]. Although multiphase and compositional models allow simulating aqueous and gaseous flow and transport in a coupled manner, they do not include a wide range of geochemical reactions, but typically focus on interphase mass transfer, with reaction capabilities limited to first-order decay and linear sorption [*Abriola and Pinder*, 1985; *Corapcioglu and Baehr*, 1987; *Falta et al.*, 1989; *Sleep and Sykes*, 1989; *Sleep*, 1998]. In a recent review of public domain codes for modeling the fate of VOC mixtures in the unsaturated zone, *Karapanagioti et al.* [2003] recognized that only a small number of codes consider vapor advection and even fewer models include reaction processes such as biodegradation.

[10] Only a limited number of models exists that couple gas transport, solute transport and geochemical reactions in the vadose zone [*Šimůnek and Suarez*, 1994; *White*, 1995; *Xu et al.*, 2000; *Mayer et al.*, 2002; *Steefel*, 2004; *Saaltink et al.*, 2004; *Lichtner et al.*, 2004; *Linklater et al.*, 2005]. Although some of these models considered gas advection [*White*, 1995; *Xu et al.*, 2000; *Saaltink et al.*, 2004; *Lichtner et al.*, 2004; *Linklater et al.*, 2005], none of these models incorporates multicomponent gas transport (i.e., DGM, Stefan-Maxwell) into a rigorous description of the feedback between geochemical reactions and gas transport processes.

[11] Here the effect of geochemical reactions on multicomponent gas and solute transport in the vadose zone is investigated through the development and application of a reactive transport model. This model is the first attempt to introduce the dusty gas model into a multicomponent reactive transport code that considers a full suite of geochemical reactions including intra-aqueous reactions and mineral dissolution-precipitation. The focus of this work is on investigating the role of reaction-induced gas transport.

The model is applied to several relevant problems including methane oxidation in unsaturated sediments and its release to the atmosphere and interactions between sulfide mineral oxidation, ingress of atmospheric oxygen, carbonate mineral dissolution, and CO_2 release to the atmosphere.

2. Model Development

[12] The present model builds on the existing reactive transport model MIN3P [*Mayer et al.*, 2002]. The model description given here focuses on the development of the equations that account for the contributions of multicomponent gas transport to the mass balance equations. Other aspects of the model are only briefly described and the reader is referred to *Mayer et al.* [2002] for details. The mass balance equation for component k present in the aqueous and gaseous phases and written in terms of total component concentrations takes the form:

$$\begin{aligned}&\frac{\partial}{\partial t}\left[S_a\phi T_a^k\right]+\frac{\partial}{\partial t}\left[S_g\phi T_g^k\right]\\&\quad+\nabla\cdot\left[\mathbf{q}_a T_a^k\right]+\nabla\cdot\left[\mathbf{q}_g T_g^k\right]-\nabla\cdot\left[S_a\mathbf{D}_a\nabla T_a^k\right]-\nabla\cdot\mathbf{N}_{T,g}^k\\&\quad-Q_{a,a}^k-Q_{a,s}^k-Q_{a,ext}^k-Q_{g,ext}^k=0\end{aligned}\tag{1}$$

where t [s] is time, ϕ is the porosity [m^3 void m^{-3} porous medium], S_a [m^3 H_2O m^{-3} void] is the saturation of the aqueous phase, and S_g [m^3 gas m^{-3} void] is the saturation of the gaseous phase. $\mathbf{q}_a$ and $\mathbf{q}_g$ [m s^{-1}] are the Darcy flux vectors in the aqueous and the gaseous phases, respectively, while $\mathbf{D}_a$ [m^2 $s{-}^1$] is the dispersion tensor for the aqueous phase, and $\mathbf{N}_{T,g}^k$ [mol m dm^{-3} porous medium s^{-1}] is the total diffusive flux vector for component k in the gaseous phase. T_a^k [mol L^{-1} H_2O] defines the total aqueous component concentration for component k, while T_g^k [mol L^{-1} gas] is the total gaseous concentration for component k. These total concentration terms implicitly include equilibrium reactions such as hydrolysis, aqueous complexation, aqueous phase oxidation reduction, and mass transfer between the aqueous and gas phases [*Mayer et al.*, 2002]. Ion exchange and surface complexation reactions can also be considered as equilibrium reactions [e.g., *Mayer et al.*, 2001a, 2001b]. On the other hand, slow intra-aqueous reactions and mineral dissolution-precipitation are described using a generalized kinetic formulation [*Mayer et al.*, 2002]. These reactions contribute to the mass balance equations through source-sink terms. $Q_{a,a}^k$ [mol dm^{-3} porous medium s^{-1}] and $Q_{a,s}^k$ [mol dm^{-3} porous medium s^{-1}] in equation (1) represent the internal sources and sinks toward the total aqueous component concentrations due to intra-aqueous kinetic reactions and kinetically controlled dissolution-precipitation reactions, respectively. A detailed description of the geochemical reaction network can be found in *Mayer et al.* [2002]. The model is database-driven [*Ball and Nordstrom*, 1991] and its flexibility to include a wide range of geochemical processes has been demonstrated previously [e.g., *Mayer et al.*, 2001a, 2001b, 2002; *Jurjovec et al.*, 2004; *Mayer et al.*, 2006]. $Q_{a,ext}^k$ [mol dm^{-3} porous medium s^{-1}] and $Q_{g,ext}^k$ [mol dm^{-3} porous medium s^{-1}] are external source and sink terms for the aqueous and gaseous phase boundary fluxes, respectively.

[13] Equation (1) has been modified from *Mayer et al.* [2002] by the addition of the term $\nabla\cdot[\mathbf{q}_g T_g^k]$ to account for gas advection and the replacement of the Fickian diffusion term in the gas phase by a multicomponent diffusion term defined by $\nabla\cdot\mathbf{N}_{T,g}^k$. Mechanical dispersion in the gas phase is not included in the model. *Popovicova and Brusseau* [1997] report for an experimental study of methane migration, that mechanical dispersion can be neglected, unless advective velocities are large (>288 m d^{-1}). Although the experiments were conducted with glass beads, characterized by medium properties that are quite different from natural sediments, the maximum velocities for the examples presented in this work (ranging from 0.0016 to 5.6 m d^{-1}) justify the omission of dispersive transport in the gas phase.

[14] Despite both advective flux terms in (1) have the same form, $\mathbf{q}_a$ and $\mathbf{q}_g$ are calculated differently. Assuming steady state flow conditions in the aqueous phase and a passive gas phase, the Darcy flux in the aqueous phase is obtained by solving the variably saturated flow problem by means of Richard's equation and Darcy's law [*Mayer et al.*, 2002]:

$$S_aS_s\frac{\partial h}{\partial t}+\phi\frac{\partial S_a}{\partial t}-\nabla\cdot\left[k_{ra}\mathbf{K}\nabla h\right]-Q_a=0\tag{2}$$

$$\mathbf{q}_a=-k_{ra}\mathbf{K}\nabla h\tag{3}$$

where S_s [m^{-1}] defines the specific storage coefficient and k_{ra} (dimensionless) is the relative permeability of the porous medium with respect to the aqueous phase, h [m] is the hydraulic head, and Q_a is a source-sink term [s^{-1}]. $\mathbf{K}$ = $(\rho_a g\mathbf{k}/\mu_a)$ [m s^{-1}] is the hydraulic conductivity tensor, with ρ_a [kg m^{-3}] being the aqueous phase density, g [m s^{-2}] being the gravitational acceleration, μ_a [kg m^{-1} s^{-1}] being the aqueous phase viscosity, and $\mathbf{k}$ [m^2] being the intrinsic permeability tensor [*Bear*, 1972]. Possible boundary conditions include specified head (Dirichlet), specified flux (Neumann), and a seepage face boundary condition.

[15] The advective flux in the gaseous phase, on the other hand, is directly substituted in the mass balance equation (1). It is calculated using Darcy's law as a function of the total pressure in the gas phase (p_g [Pa])

$$\mathbf{q}_g=-\frac{k_{rg}\mathbf{K}}{\mu_g}\left(\nabla p_g+\rho_g g\nabla z\right)\tag{4}$$

where k_{rg} (dimensionless) is the relative permeability of the porous medium with respect to the gaseous phase (Appendix A, (A4)), ρ_g [kg m^{-3}] and μ_g [kg m^{-1} s^{-1}] are the gaseous phase density and viscosity, respectively. Gas density and viscosity are calculated as a function of the gas composition according to equations (A1) and (A2) (see Appendix A). Total pressure in the gas phase is related to the sum of the partial pressures of all gas species (p_g^i [Pa]):

$$p_g=\sum_{i=1}^{N_g}p_g^i\tag{5}$$

By means of the ideal gas law, partial pressures relate to the molar concentrations of gas species (c_g^i [mol L^{-1} gas]):

$$p_g^i=RTc_g^i\tag{6}$$

with R [J mol^{-1} K^{-1}] being the ideal gas law constant, and T [K] the temperature of the system. For the range of pressures and temperatures usually encountered in the shallow subsurface environment, the use of the ideal gas law for real gases is not expected to lead to significant errors [*Falta et al.*, 1989].

[16] Since concentrations of the gas species are obtained from the solution of the reactive transport equation (1), gas phase velocities adjust to compositional changes in the gas phase. In contrary to the formulation used for advection in the aqueous phase, advection in the gas phase is fully coupled with the processes that modify gas concentrations. This approach ensures that a direct link between gas transport processes and chemical reactions is provided: any change in gas concentrations caused by chemical reactions has a direct effect on the gradient that drives gas transport, as described by equations (4), (5), and (6).

[17] First type boundary conditions can be applied in the aqueous phase (specified concentrations) or in the gas phase (specified partial pressures). In addition, second type boundary conditions (specified fluxes) can be defined separately for the aqueous and gas phases; and a free exit boundary condition can be used for the aqueous phase.

[18] Diffusive fluxes in the gas phase can be calculated using the Stefan-Maxwell equations within the framework of the dusty gas model [*Mason and Malinauskas*, 1983]:

$$\sum_{\substack{j=1\\ j\neq i}}^{N_g} \frac{\mathbf{N}_g^i \chi_g^j - \mathbf{N}_g^j \chi_g^i}{S_g \phi \tau_g D_g^{ij}} + \frac{\mathbf{N}_g^i}{D_{K,g}^i} = -\nabla c_g^i - \frac{M_g^i c_g^i g}{RT} \nabla z \qquad i = 1, \ldots, N_g \tag{7}$$

where χ_g^i (dimensionless) is the molar fraction of gas species i, M_g^i [kg mol^{-1}] is the molecular weight of gas species i, and $\mathbf{N}_g^i$ [mol m dm^{-3} porous medium s^{-1}] is the diffusive molar flux of gas species i. The right hand side of (7) must be viewed as the driving force for diffusive transport of gas species i [*Thorstenson and Pollock*, 1989], which includes the gravitational field in the second term [*Sleep*, 1998]. The driving force is balanced by the friction between gas species i and all other gas species and the sediment particles as expressed by the left hand side of (7). The species concentrations and diffusive fluxes can be related to the components by means of the stoichiometric coefficients (v_g^{ik}) [*Mayer et al.*, 2002]:

$$T_g^k = \sum_{i=1}^{N_g} v_g^{ik} c_g^i \tag{8}$$

$$\mathbf{N}_{T,g}^k = \sum_{i=1}^{N_g} v_g^{ik} \mathbf{N}_g^i \tag{9}$$

In equation (7), the first term on the left hand side accounts for molecule-molecule interactions, with D_g^{ij} [m^2 s^{-1}] being the free phase binary diffusion coefficient between gas species i and j, and τ_g the gas tortuosity coefficient (see equations (A6) and (A7)). The second term accounts for molecule-sediment interactions, with $D_{K,g}^i$ [m^2 s^{-1}] being the Knudsen diffusion coefficient for species i. The Knudsen diffusion coefficient can be calculated as a function of the Klinkenberg parameter (b_i [Pa]) [*Thorstenson and Pollock*, 1989; *Massmann and Farrier*, 1992; *Sleep*, 1998; *Fen and Abriola*, 2004]:

$$D_{K,g}^i = \frac{k_{rg} k}{\mu_g^i} b_i = \frac{k_{rg} k}{\mu_g^i} \left(b_{ref} \frac{\mu_g^i}{\mu^{ref}} \sqrt{\frac{M^{ref}}{M_g^i}} \right) \tag{10}$$

where M_{ref} [Kg mol^{-1}] is the molecular weight of the gas of reference. The Klinkenberg parameter of the reference gas (b_{ref}) can be measured or estimated using expressions (A8) or (A9).

[19] When $D_g^{ij} \gg D_{K,g}^i$, the first term in equation (7) becomes negligible, and the system is said to be in the Knudsen flow regime. In this regime, gas molecules do not interact with each other but only with the sediment particles. On the other hand, when gas molecules interact only with other gas molecules ($D_g^{ij} \ll D_{K,g}^i$), the system is in the molecular regime and equations (7) can be simplified to the basic form of the Stefan-Maxwell equations [*Thorstenson and Pollock*, 1989]:

$$\sum_{\substack{j=1\\ j\neq i}}^{N_g} \frac{\mathbf{N}_g^i \chi_g^j - \mathbf{N}_g^j \chi_g^i}{S_g \phi \tau_g D_g^{ij}} = -\nabla c_g^i - \frac{M_g^i c_g^i g}{RT} \nabla z \qquad i = 1, \ldots, N_g \tag{11}$$

In this form, equations (11) are linearly dependent. To make them a linearly independent set of equations, one equation in (11) needs to be replaced by an additional constraint. For example, the constraint of equimolar counterdiffusion can be used, which states that the sum of diffusive fluxes of all species is zero:

$$\mathbf{N}_g^D = \sum_{i=1}^{N_g} \mathbf{N}_g^i = 0 \tag{12}$$

In general, however, the so-called nonequimolar flux ($\mathbf{N}_g^D$) is not zero. The nonequimolar flux occurs in systems with walls (soil particles) because diffusing species have different molecular weights: lighter molecules move faster than heavier molecules [*Cunningham and Williams*, 1980]. Although diffusive in origin, the nonequimolar flux results in a nonseparative flux in the same direction as the diffusive flux of the lightest gas species in the mixture. In conclusion, the diffusive flux of gas species i ($\mathbf{N}_g^i$) can be split in two components: a separative flux ($\mathbf{J}_g^i$) and a nonseparative flux ($\chi_g^i \mathbf{N}_g^D$) [*Thorstenson and Pollock*, 1989] such that

$$\mathbf{N}_g^i = \mathbf{J}_g^i + \chi_g^i \mathbf{N}_g^D \tag{13}$$

The use of the constraint expressed by (12) implies that the nonequimolar component of the diffusive fluxes is not included ($\mathbf{N}_g^i = \mathbf{J}_g^i$) [*Fen*, 1993]. For a nonequimolar formulation of equations (11), the additional constraint can be set by Graham's law of effusion/diffusion [*Fen*, 1993]:

$$\sum_{i=1}^{N_g} \left(M_g^i \right)^{1/2} \mathbf{N}_g^i = 0 \tag{14}$$

If one adds the DGM equation for each gas species (7) over all gas species, the first term on the left hand side cancels out, and the right hand side can be rearranged as a function of bulk phase properties using equations (5) and (A1):

$$\sum_{i=1}^{N_g} \frac{\mathbf{N}_g^i}{D_{K,g}^i} = -\frac{1}{RT}\left(\nabla p_g + \rho_g g \nabla z\right) \tag{15}$$

Having in mind that Knudsen diffusion coefficients are inversely proportional to the square root of the molecular weights of the gas species (equation (10)), comparison of equations (14) and (15) shows that the DGM formulation implicitly includes the nonequimolar flux component. Further, in the isobaric case ($\nabla p_g + \rho_g g \nabla z = 0$), equation (15) simplifies to (14).

[20] Finally, when D_g^{ij} and $D_{K,g}^i$ have similar magnitudes, the system is in the so-called transition regime, and both free phase and Knudsen diffusion processes contribute to the diffusive fluxes.

3. Numerical Implementation

[21] The different formulations presented in section 2 have been implemented in a modular fashion into an existing computer code for the simulation of multicomponent reactive transport problems in variably saturated porous media [*Mayer et al.*, 2002]. The resulting code now allows the user to select the model of gas diffusion: the Stefan-Maxwell equations in the form of the DGM (7), the basic Stefan-Maxwell equations (11), or the standard Fickian description (as implemented by *Mayer et al.* [2002]). The model also includes a gas advection term (4).

[22] The domain is discretized spatially using the finite volume method, and an implicit scheme is employed for time discretization. To increase the efficiency of the solution, avoid oscillations, and to ensure its physical correctness when sharp concentration fronts are present, an upstream weighting scheme has been employed for solving the advective component of the transport equations. Although this standard Eulerian formulation can lead to poor accuracy when simulating the advective-diffusive migration of concentration fronts [*Daus et al.*, 1985], the systems investigated in this work are characterized by rapid gas transport under quasi-steady state conditions with concentration gradients mostly controlled by boundary conditions and geochemical reactions. Under these conditions, Eulerian methods provide adequate accuracy, even for large time steps [*Saaltink et al.*, 2001]. The implementation is unconditionally stable, and in the simulations presented in the next section, no stability issues have appeared.

[23] The concentrations of the components as species in solution (c_a^i) have been chosen as the primary variables [*Mayer et al.*, 2002]. Concentrations of aqueous complexes, gases, surface complexes and secondary redox species in equilibrium are dependent variables [*Mayer et al.*, 2002].

[24] In the current implementation, the diffusive flux equations ((9), (7), and (11)) and the advective flux equations ((4), (5), and (6)) are substituted directly into the mass balance equations (1). Although there is no consensus about the relative merits of the various coupling approaches for reactive transport problems [e.g., *Bell and Binning*, 2002], some authors have favored the direct substitution approach (DSA) over sequential approaches for problems involving a high number of flushed pore volumes. The DSA allows for the use of relatively large time steps without loss of accuracy, while the use of sequential approaches in such systems often requires small time steps leading to long computation times [*Saaltink et al.*, 2001; *Calderhead and Mayer*, 2004]. In the case of gas transport, low viscosities of soil gas and large diffusion coefficients lead to short transport timescales, and the solution of the governing equations can benefit from the direct substitution approach, as implemented here.

[25] The code uses an adaptive time stepping scheme, which determines the size of the time increment based on the relative change in aqueous concentrations and the number of Newton iterations in the previous time step. The size of the time steps is limited by a user-specified maximum value or, if smaller than that value, by the stiffness of the set of equations. In the systems of interest, stiffness originates in the fact that the processes involved have very different timescales: rapid gas transport compared to relatively slow reactions.

4. Gas Transport Mechanisms and Model Verification

[26] The model was verified by comparison to literature examples of gas transport modeling. These examples are also used to highlight the capabilities of the code including transport in binary and multicomponent mixtures by molecular diffusion, Knudsen diffusion and advection. A validation of the model's capability to couple gas transport and geochemical reactions is provided by the example of methane oxidation in landfill cover soils. The geochemical module has been verified previously and has been applied to a number of studies [e.g., *Mayer et al.*, 2001a, 2001b, 2002, 2006].

4.1. Nonreactive Binary Transport in the Molecular and Transition Regimes

[27] The model presented in the previous section can simulate advective-diffusive transport in both molecular and transition regimes. The results obtained with the DGM presented by *Fen and Abriola* [2004] (hereinafter referred to as FA) for the transport of N_2 and CH_4 in a 1-D horizontal domain under three different sets of conditions are compared to results obtained using the present DGM formulation (equation (7)). Simulation parameters are presented in Table 1. For all cases, the pore space is initially filled with N_2. At time zero, the methane mole fraction is set to 1 at the left boundary. In the absence of an imposed pressure gradient, diffusion-dominated transport occurs in the molecular regime when a high-permeability porous medium is considered (case 1, Figure 1a). The model is also able to simulate Knudsen diffusion in low-permeability media (case 2, Figure 1b), which results in reduced diffusive fluxes with respect to those in the molecular regime (Figure 1a). On the other hand, advection-dominated transport occurs under an imposed pressure gradient of 10 Pa m^{-1} that drives advective flux away from the methane source (case 3, Figure 1c). For all cases, the results obtained using the DGM implemented in the present model agree well with results obtained by FA (Figure 1).

Table 1. Summary of Parameters and Equations Used in Section 4

	Fen and Abriola [2004]			*Thorstenson and Pollock* [1989], Figures 2 and 3	*De Visscher and Van Cleemput* [2003]	
	Figure 1a	Figure 1b	Figure 1c		Figures 4a and 5	Figure 4b
Porosity ϕ, m^3 void m^{-3} porous medium	0.4	0.4	0.4	1.0	0.5878	0.5878
Gas saturation S_g, m^3 gas m^{-3} void	1.0	1.0	1.0	1.0	0.7	0.7
Permeability k, m^2	10^{-10}	10^{-16}	10^{-10}	10^{-12}	2×10^{-13}	2×10^{-13}
Diffusion equations	(7)	(7)	(7)	(7)	(11, 12)	(7)
Gas tortuosity τ_g	0.736 (A6)	0.736 (A6)	0.736 (A6)	0.1	0.1854 (A7)	0.1854 (A7)
Klinkenberg parameter b_{ref}, Pa	(A8)	(A8)	(A8)	$b_{N2} = 1.35 \cdot 10^4$	N/A	(A9)
Reaction rate constant, mol L^{-1} H_2O s^{-1}	N/A	N/A	N/A	N/A	4.43×10^{-6}	
Grid size, m	0.047	0.047	0.047	0.62	0.026	0.026

4.2. Nonreactive Multicomponent Transport in the Molecular Regime

[28] In addition to binary transport, the present model incorporates multicomponent diffusion and advection. Example 1 of *Thorstenson and Pollock* [1989] (hereinafter TP) presents the simulation of steady state multicomponent gas transport in a 10-m column of unsaturated porous medium with a permeability of 10^{-12} m^2 (Table 1). Whereas the top boundary is kept at atmospheric conditions ($p_g^{N_2}$ = 0.78, $p_g^{O_2}$ = 0.20, $p_g^{CO_2}$ = 0.01, p_g^{Ar} = 0.01, $p_g^{CH_4}$), methane is being produced at the bottom at a rate of 10^{-4} mol m^{-2} s^{-1}. In TP, the effects of gravity on gas transport are included; however, gravity gradients are not reported individually but lumped together with pressure gradients:

$$\nabla p_g' = \nabla p_g + \rho_g g \nabla z \tag{16}$$

[29] Results from the present DGM model compare well with those of TP in terms of concentrations (Figure 2), fluxes (Figure 3) and gradients ($\nabla p_g'$ = 24.4 Pa m^{-1}). The implementation of the DGM also allows accounting for the nonequimolar flux component of diffusion. Methane, which constitutes most of the gas phase in the column, has a smaller molecular weight than gases in the atmosphere, and causes an upward nonequimolar component in the diffusive fluxes (Figure 3).

4.3. Methane Attenuation and Multicomponent Transport in Landfill Cover Soils

[30] In order to study the capacity of a landfill cover soil to attenuate the emissions of contaminant gases to the atmosphere, *De Visscher and Van Cleemput* [2003] conducted a column experiment with artificial landfill gas. A constant flux of CH_4 (13.4 mol m^{-2} d^{-1}) and CO_2 (13.4 mol m^{-2} d^{-1}) was applied at the bottom of the column while the top was kept at atmospheric conditions ($p_g^{N_2}$ = 0.78, $p_g^{O_2}$ = 0.21, $p_g^{CO_2}$ = 0.01, $p_g^{CH_4}$ = 0.0). The column was microbially reactive after being filled with landfill top cover soil and methane was oxidized by atmospheric oxygen diffusing into the column.

[31] The capability of the model to couple a wide range of geochemical reactions, as well as gas and solute transport makes it adequate to simulate these experiments. Simulation results are compared to experimental data and to modeling results by *De Visscher and Van Cleemput* [2003] (hereinafter DV), who developed a model based on the Stefan-Maxwell equations specifically designed to simulate the experimental results. Of two possible stoichiometric relationships for the oxidation of methane in the column

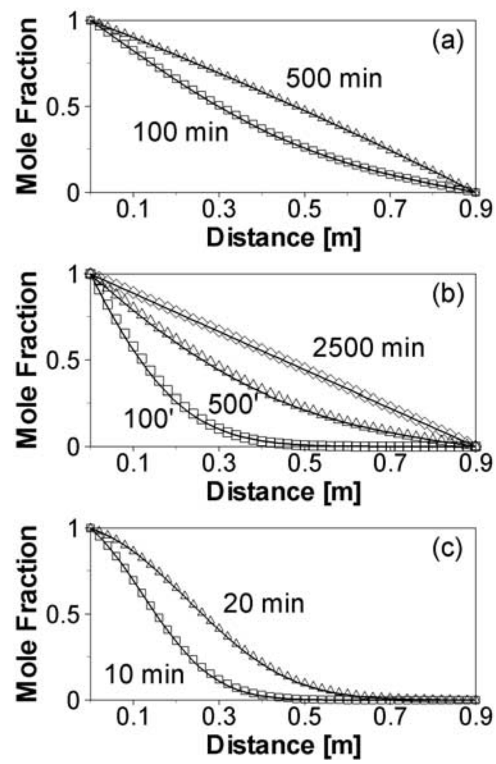

Figure 1. Methane concentrations as a result of (a) diffusion in the molecular regime, (b) diffusion in the transition regime, and (c) advection-dominated transport in the molecular regime. Model results are solid lines; results from *Fen and Abriola* [2004] are shown as symbols.

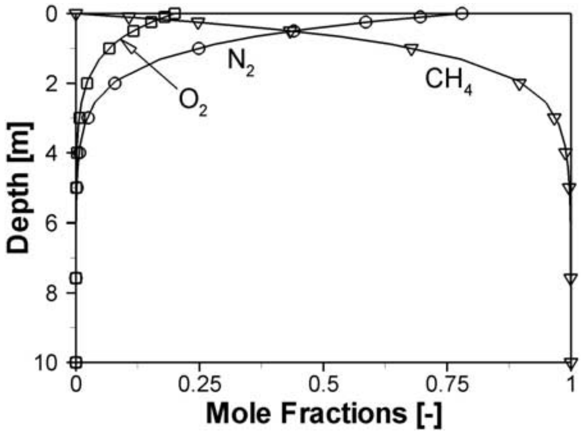

Figure 2. Comparison between model results (solid lines) and results from *Thorstenson and Pollock* [1989] (symbols) for the concentrations of N_2, O_2, and CH_4.

explored in DV, equation (17) provided a better fit to experimental data, and is used in this work:

$$CH_4 + 1.5O_2 \rightarrow 0.5CO_2 + 1.5H_2O + 0.5CH_2O \qquad (17)$$

In DV, the reaction rate was modeled with a kinetic expression that accounted for variable methanotrophic activity. However, it was found that model results were relatively insensitive to it. Therefore, in the present work the following hyperbolic expression is used:

$$R_{CH_4} = -k_{CH_4}\left(\frac{C_a^{CH_4}}{K_1 + C_a^{CH_4}}\right)\left(\frac{C_a^{O_2}}{K_2 + C_a^{O_2}}\right) \qquad (18)$$

where K_1 and K_2 are the half-saturation constants for CH_4 and O_2, respectively. The reaction rate constant (k_{CH_4}) is set equal to 7.5×10^{-7} moles $\text{kg}^{-1}{}_{\text{DRY WEIGHT}}$ s^{-1} in accordance with the maximum oxidation rate reported by DV. Permeability was not reported in DV, and it is estimated at 2×10^{-13} m^2 corresponding to a loamy sand. The parameter values used in the simulation are summarized in Tables 1 and 2.

[32] Model results obtained with the present model (using the equimolar Stefan-Maxwell equations (11) and (12)) are shown in Figure 4a together with DV's model results and experimental data [*De Visscher and Van Cleemput*, 2003]. The present model produces almost identical results to DV's model; small discrepancies encountered mostly near the reaction zone can be attributed to the slightly different reaction rate model used that leads to differences in the calculated reaction rates of less than 3% with respect to those of DV's model (not shown). Results obtained using the DGM (7) (Figure 4b) are slightly different than those obtained using the equimolar Stefan-Maxwell equations ((11) and (12)) because the DGM includes a nonequimolar flux component as well as Knudsen diffusion. A more detailed comparison between the two formulations is beyond the scope of this paper; the reader is referred to other works [*Massmann and Farrier*, 1992; *Sleep*, 1998; *Fen and Abriola*, 2004]. It can be observed that the oxidation of methane leads to a decrease in pressure gradient in the reaction zone, which in turn results in a decrease in the advective velocity (Figure 5a). As a result, advection is not a relevant process for methane emission to the atmosphere (Figure 5b).

5. Investigation of Interactions Between Transport and Reactions

[33] In this section, the interactions between multicomponent gas transport, solute transport, and geochemical reactions are investigated for two systems of relevance in environmental earth sciences. Special attention is paid to the effects of reactions on gas advection, and multicomponent gas diffusion as well as the effects of interphase mass transfer processes on the composition of the gas phase. Aqueous phase flow and transport is also included and flow is assumed at steady state; moisture content thus remains fixed over the simulation time. In addition, atmospheric pressure fluctuations are excluded. Such fluctuations are superimposed on the reaction-induced behavior, but it is assumed that their effects average out over time, while reaction induced transport is a unidirectional process. The DGM is used to calculate diffusive fluxes for both examples, as it provides the most comprehensive representation of gas diffusion in porous media.

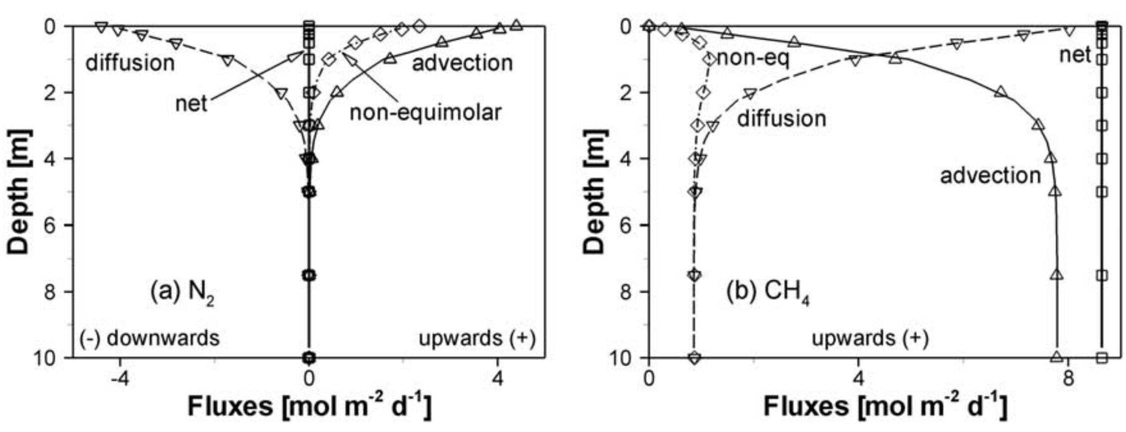

Figure 3. Advective, diffusive, and nonequimolar components of the net flux for (a) N_2 (g) and (b) CH_4 (g). Model results are solid lines; results from *Thorstenson and Pollock* [1989] are shown as symbols.

Table 2. Viscosity and Binary Diffusion Coefficients Used in the Simulations in Sections 3 and 4[a]

	Viscosity $\mu_g{}^i$, Pa s	Binary Diffusion Coefficients D_g^{ij}, $m^2\ s^{-1}$ O_2	CO_2	CH_4	Ar
N_2	1.76×10^{-5}	2.083×10^{-5}	1.649×10^{-5}	2.137×10^{-5}	1.954×10^{-5}
O_2	2.05×10^{-5}		1.635×10^{-5}	2.263×10^{-5}	1.928×10^{-5}
CO_2	1.47×10^{-5}			1.705×10^{-5}	1.525×10^{-5}
CH_4	1.09×10^{-5}				2.045×10^{-5}
Ar	2.23×10^{-5}				

[a]T = 298 K.

5.1. Production and Consumption of Methane at a Crude Oil Spill Site

[34] Gas data collected by *Amos et al.* [2005] to study the natural attenuation of petroleum hydrocarbons at an oil spill site near Bemidji, Minnesota, suggest that biodegradation reactions cause gas advection in unsaturated contaminated sediments. Directly above the water table, measured concentrations of nonreactive gases such as argon and at this site also nitrogen, were lower than atmospheric levels, suggesting that advective gas transport toward the ground surface takes place. It was hypothesized that biodegradation by methanogenesis in the zone near the water table may cause an increase of the gas pressure driving gas advection upward, which would explain the decrease of nitrogen and argon pressures in this zone (Figure 6). However, higher above the water table, a region of enrichment of Ar and N_2 was observed relative to atmospheric composition. This zone coincides with a region of methanotrophic activity, i.e., the oxidation of methane by oxygen. These data suggest that the decrease in gas pressure in the reaction zone drives advective gas transport into this zone. Results obtained in the landfill cover soil example (section 4.3) also suggested that the decrease in gas pressure caused by methane oxidation affects advective and diffusive fluxes (Figures 5a and 5b). Bemidji field data shows that while gases consumed in the reaction become depleted in the methanotrophic zone, gases that do not participate in the reaction such as nitrogen and argon become enriched due to nonseparative transport processes into the reaction zone.

[35] Here this conceptual model is tested in a simplified manner using 7 components (O_2(aq), CH_4(aq), CO_2(aq), Ar(aq), N_2(aq), Ca^{2+}, and H^+), 5 gases (O_2(g), CH_4(g), CO_2(g), Ar(g), and N_2(g)), 2 aqueous carbonate complexes (HCO_3^-, and CO_3^{2-}), and the mineral phase calcite. For the simulations, it is assumed that pore water is in equilibrium with calcite, because the native aquifer material contains approximately 4–6% carbonates [*Bennett et al.*, 1997]. The system is approximated using a one-dimensional vertical section, in which CH_4 and CO_2 are produced at the bottom due to anaerobic degradation of the oil. Methane oxidation along the section is simulated using the rate expression given in equation (18) and according to the following stoichiometry:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \tag{19}$$

[36] The measured concentration profiles indicate significant concentration changes around 2–2.5 m of depth (Figure 6), which suggests the presence of a zone with inhibited gas mobility, possibly due to increased water content. The location of this zone coincides with a layer of fine-grained sediments observed in core samples from the site near the location of well 601 (USGS, unpublished data, 1987). In the simulation, a layer of lower permeability (10^{-13} m s^{-1}) has been considered between 2.0 and 2.5 m depth, while it is 2×10^{-12} m s^{-1} above and 5×10^{-12} m s^{-1} below. A recharge rate of 160 mm yr^{-1} results in the simulated aqueous saturation profile shown in Figure 7. Porosity is 0.30 in the finer grained zone and 0.38 elsewhere; aqueous advection and

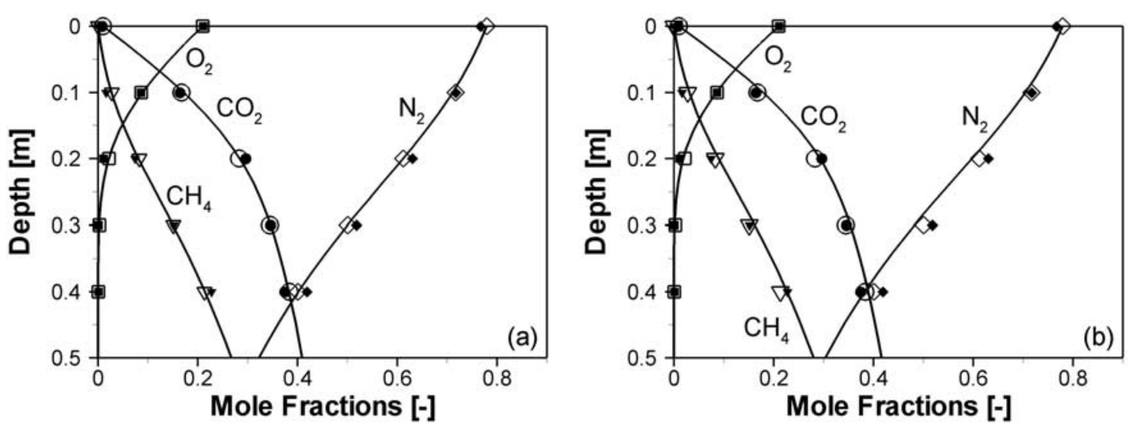

Figure 4. Comparison between model results (solid lines) and results from *De Visscher and Van Cleemput* [2003] (open symbols) for the concentrations of N_2, O_2, CO_2, and CH_4. Solid symbols are experimental data. (a) Stefan-Maxwell equations and (b) dusty gas model.

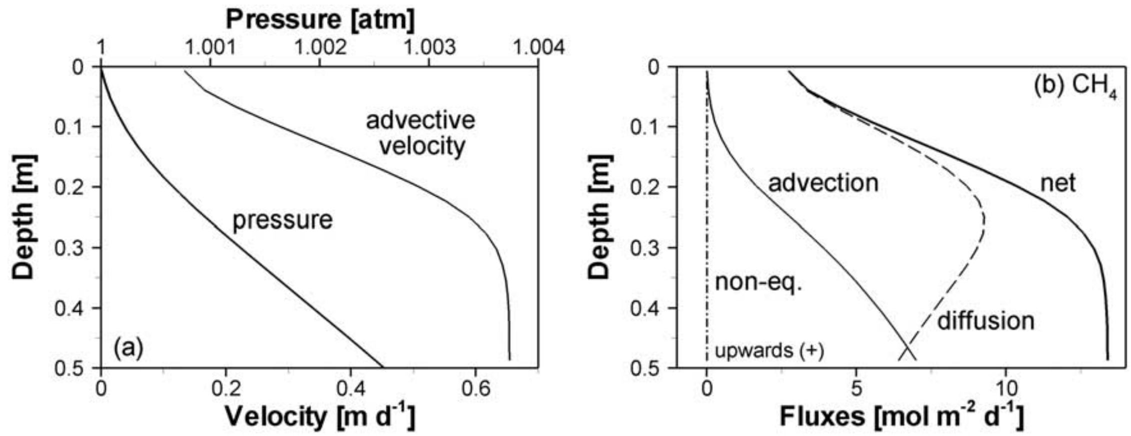

Figure 5. (a) Velocity and pressure in the landfill cover soil column simulated using the equimolar Stefan-Maxwell equations. (b) Advective, diffusive, and nonequimolar components of the net flux of CH_4 simulated using the equimolar Stefan-Maxwell equations.

dispersion are included. A summary of model parameters is given in Table 3.

[37] The influx of CH_4 and CO_2 at depth and the rate of methane oxidation within the sediment column are used as calibration parameters. The parameter values that provide good agreement to observed data are 0.09 mol m^{-2} d^{-1} and 0.009 mol m^{-2} d^{-1} for the influxes of CH_4 and CO_2, and 10^{-7} moles L^{-1} s^{-1} for the reaction rate (k_{CH_4}) respectively. The relatively high CH_4 to CO_2 ratio may be attributed to a range of processes including the preferential release of CH_4 from the saturated zone by ebullition [*Amos and Mayer*, 2006]; the reduced nature of carbon in crude oil [*Hunt*, 1979], and sequestration of CO_2 in the sediments, e.g., by the precipitation of siderite ($FeCO_3$) [*Mayer et al.*, 2002]. The simulations intend to represent the current steady state conditions at the oil spill site. Results are shown at 3.17 years, at which time steady state was reached when the initial and boundary conditions provided in Table 4 were used.

[38] The reasonable agreement between observed and simulated data that could be obtained (Figure 6) supports the validity of the conceptual model proposed by *Amos et al.* [2005]. As a result of methane oxidation (equation (19)), 3 moles of gases (CH_4 and O_2) are transformed into 1 mol of CO_2. As recognized by *Scheutz and Kjeldsen* [2003], this transformation causes a decrease in total gas pressure in the reaction zone. At the Bemidji site, a reversal of the gradient ($\nabla p'_g$) is observed in the reaction zone: it is upward in the bottom 3 m (0.21 Pa m^{-1}), and downward in the top 2.5 m (0.55 Pa m^{-1}). This causes a reversal of the advective fluxes in the upper part of the section (Figure 8). Atmospheric O_2 is supplied at the top of the section not only by diffusion (92.6%) but also by advection (7.4%) (Figure 8d). The N_2 concentration profile in Figure 6 can only be explained by considering this downward advective component. The concentration gradient drives diffusive fluxes away from the reaction zone (upward above 2.77 m and downward below 2.77 m). As it is assumed that N_2 is nonreactive, the net flux of N_2 at steady state must be zero; therefore an advective flux of the same magnitude as the diffusive flux but in opposite direction must develop (Figure 8a). The nonequimolar N_2 flux component also contributes to counteract diffusion except in a very thin layer near 2.75 m where it has the same direction as the diffusive flux (Figure 8a).

[39] Comparing these results with those obtained for the landfill cover soil simulation, one can observe that at the Bemidji site the emission of methane to the atmosphere is effectively shut off by both methane oxidation and reversal

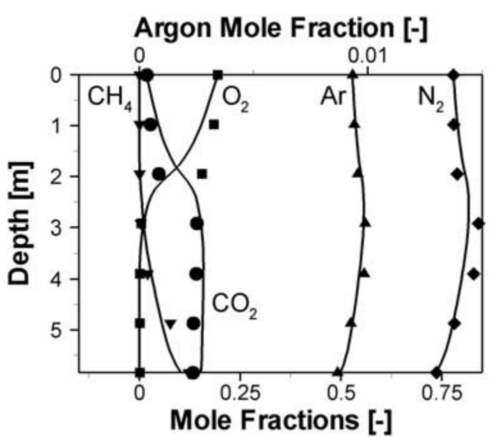

Figure 6. Molar fractions of O_2, CO_2, CH_4, N_2, and Ar along the section. Solid lines are model results; symbols are field data. A relative enrichment of N_2 and Ar and depletion of atmospheric O_2 is observed around the reaction zone.

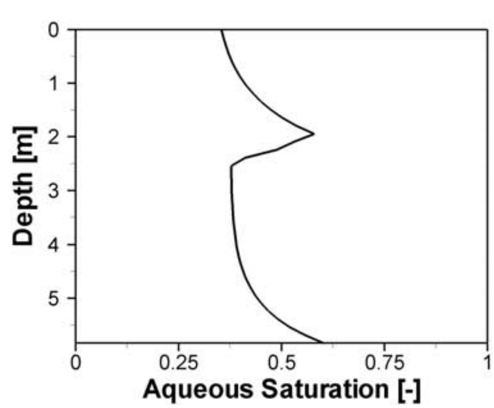

Figure 7. Simulated aqueous saturation profile.

Table 3. Summary of Parameters and Models Used in Section 5

	Bemidji[a]	Acid Mine Drainage[b]
Porosity ϕ, m^3 void m^{-3} porous medium	0.30–0.38	0.50
Gas saturation S_g, m^3 gas m^{-3} void	0.4–0.6	0.0–0.5
Permeability k, m^2	10^{-13} -2×10^{-12} $-$ 5×10^{-12}	10^{-14}
Diffusion equations	(7)	(7)
Gas tortuosity τ_g	*Millington* [1959], (A6)	*Millington* [1959], (A6)
Aqueous dispersivity, m	0.1	0.0
Klinkenberg parameter b_{ref}, Pa	*Heid et al.* [1950], (A8)	*Heid et al.* [1950], (A8)
Initial mineral volume fractions, m^3 mineral m^{-3} bulk	0.06 (calcite)	0.04 (pyrite), 0.3 (calcite), 0.0 (ferrihydrite), 0.0 (gypsum)
Reaction rate constant, mol L^{-1} s^{-1}	10^{-7} (methane oxidation)	10^{-8} (pyrite), 10^{-7} (calcite), 10^{-7} (ferrihydrite), 10^{-8} (gypsum)
Solution time, years	3.17 (steady state)	50
Grid size, m	0.15	0.05
Maximum Courant number (dimensionless)	1.39	14.34

[a]Figures 8, 9, and 10a. Values also apply to simulation results shown in Figure10b except for the initial calcite volume fraction (0 m^3 calcite m^{-3} bulk).
[b]Figures 11a, 12a, and 13a. Values also apply to simulation results shown in Figure 11b, Figure 12b, and Figure 13b except for the initial calcite volume fraction (0 m^3 calcite m^{-3} bulk).

of gas flow (Figure 8c). On the other hand, in the landfill cover example, there is a significant reduction in gas flow in the top 20 cm of the column; however, the pressure gradient is not completely reversed (Figure 5a), and methane emissions are not completely averted (Figure 5b). This implies that methane oxidation contributes to the reduction of methane emissions to the atmosphere directly by removing methane and indirectly by decreasing gas advection (landfill cover soil, Figure5b) or by completely shutting it off (Bemidji, Figure 8c). These results indicate that the significant difference in the magnitude of the gradients generated to accommodate the influxes of CH_4 and CO_2 in the landfill cover soil column (655 Pa m^{-1} or 6.5 mbar m^{-1}) and at the Bemidji site (0.21 Pa m^{-1} or 2×10^{-3} mbar m^{-1}) may be responsible for the different transport regimes in these systems, both affected by methane oxidation.

[40] At the Bemidji site, the production of CO_2 by oil biodegradation as well as by methane oxidation causes acidity to increase. However, the presence of calcite maintains pH values at around 6.5 in the lower part of the section (Figure 9a) with bicarbonate being the most abundant carbonate species (Figure 9a). Simulated pH values are in good agreement with those measured in lysimeter samples collected at the Bemidji site (pH = 6.7) (Figure 9a). If calcite is not included, simulated pH drops to values below 5 (Figure 9b). Although the simulated low pH conditions are unrealistic, this simplification would have little effect on the gas transport regime, because the dissolution of calcite contributes mostly to increase the concentration of bicarbonate but not CO_2 (compare Figures 9a and 9b). Lysimeter data also show that aqueous concentrations of Ca^{2+} increase with depth to a maximum of 3.3×10^{-3} mol L^{-1} H_2O, which is again in agreement with simulated concentrations of Ca^{2+} (Figure 9a).

[41] Aqueous flow does not appear to contribute significantly to the transport of chemical components into the column. For example, the net aqueous flux of total inorganic carbon (TIC) in the recharge water into the column is less than 1% of the net flux of CO_2(g) out of the column. The relevance of aqueous fluxes of the other gases is even smaller as their solubility is significantly smaller.

5.2. Acid Mine Drainage Generation and Attenuation

[42] A hypothetical example of acid mine drainage generation and attenuation is studied using a 2-m one-dimensional variably saturated column that initially contains pyrite with a volume fraction of 0.04. Atmospheric oxygen that is transported into the column leads to the oxidation of the pyrite present according to the following stoichiometry:

$$FeS_2 + \frac{7}{2}O_2 + H_2O \rightarrow Fe^{2+} + 2SO_4^{2-} + 2H^+ \tag{20}$$

A simple irreversible dissolution model has been employed for the reaction rate:

$$R_{FeS_2} = \min\left\{-k_{FeS_2}\left(1 - \frac{\mathrm{IAP}}{K_{FeS_2}}\right), 0\right\} \tag{21}$$

Table 4. Initial and Boundary Conditions in the Simulation of Methane Attenuation in the Crude Oil Spill Site

	Recharge Water (Atmosphere)	Groundwater		Influxes at Lower End of Column[a]
		Background	Methanogenic Zone	
Ca^{2+}	10^{-4} mol L^{-1}	10^{-4} mol L^{-1}	10^{-4} mol L^{-1}	-
N_2	0.778 atm	0.778 atm	0.729 atm	-
O_2	0.194 atm	0.194 atm	0.018 atm	-
CO_2	0.018 atm	0.018 atm	0.149 atm	0.009 mol m^{-2} d^{-1}
CH_4	0.000 atm	0.000 atm	0.111 atm	0.09 mol m^{-2} d^{-1}
Ar	0.009 atm	0.009 atm	0.009 atm	-
pH	7.0	7.0	6.5	-

[a]Only gas fluxes are reported. The lower end of the column is a free exit boundary for dissolved species.

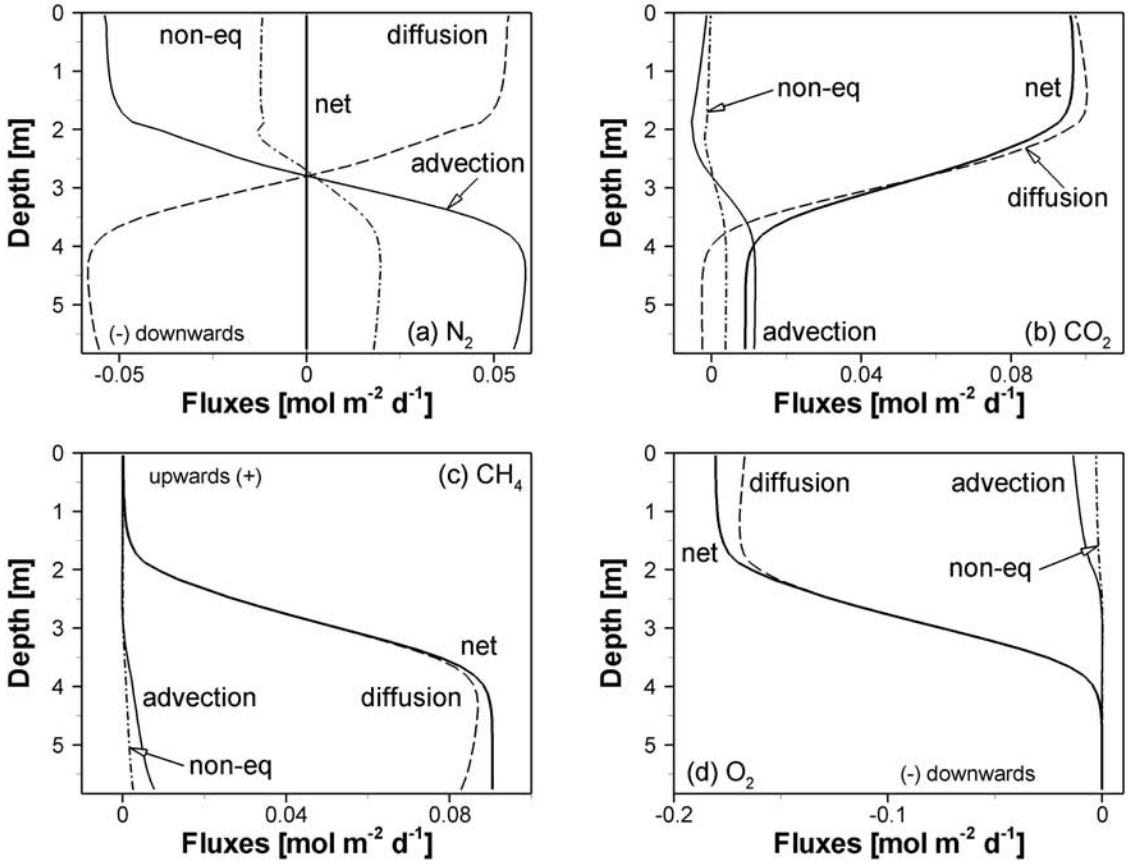

Figure 8. Advective, diffusive, nonequimolar components of the net flux for (a) N_2, (b) CO_2, (c) CH_4, and (d) O_2.

where $k_{FeS_2} = 10^{-8}$ mol dm^{-3} bulk s^{-1}, which is consistent with literature rate data for pyrite oxidation [*Williamson and Rimstidt*, 1994; *Gleisner et al.*, 2006] for surface areas found in fine grained mine waste materials. This rate coefficient leads to conditions under which the rate of pyrite oxidation is limited by the rate of oxygen supply into the reaction zone rather than by the reaction kinetics, as is often observed in mine tailings environments [*Jaynes et al.*, 1984; *Blowes et al.*, 1991; *Elberling et al.*, 1994]. IAP is the ionic activity product of the species in the reaction (20) and K_{FeS_2} is the equilibrium constant. Consumption of oxygen by pyrite oxidation (20) creates a decrease in the O_2 concentration that drives diffusive transport of O_2 into the reaction zone. In addition, the depletion of oxygen also leads to a reduction in total gas pressure that drives advective transport of atmospheric gases toward the reaction zone [*Wisotzky*, 1994; *Andersen et al.*, 2001]. The water table represents a no-flow boundary for gas phase transport

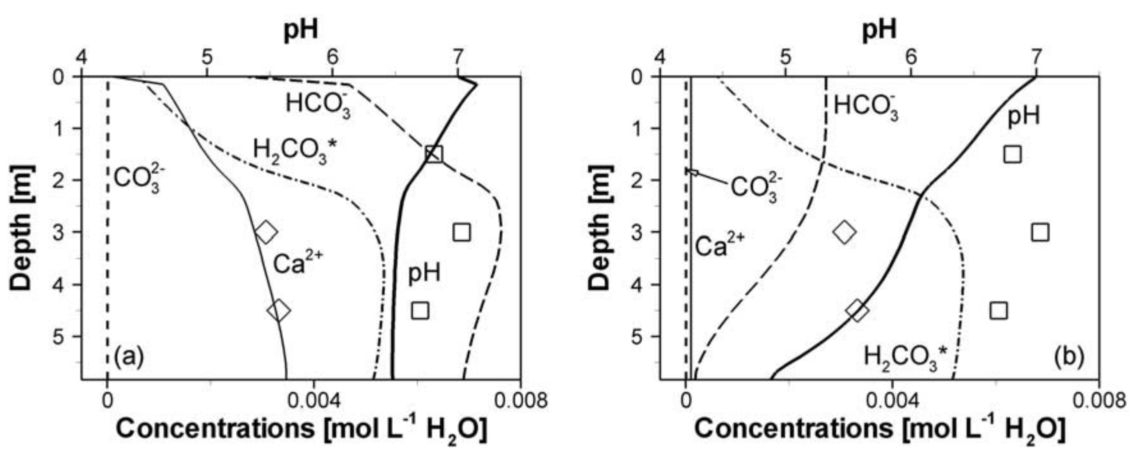

Figure 9. The pH and concentrations of Ca^{2+} and of the carbonate aqueous species (a) in the presence of calcite and (b) in the absence calcite. Lysimeter data measured near well 601 are shown as symbols: pH (squares) and Ca^{2+} (diamonds).

Table 5. Initial and Boundary Conditions in the Simulation of Pyrite Oxidation[a]

	Recharge Water (Atmosphere)	Tailings Water
Ca^{2+}, mol L^{-1}	1.00×10^{-4}	1.00×10^{-4}
Fe^{2+}, mol L^{-1}	2.03×10^{-13}	1.00×10^{-4}
Fe^{3+}, mol L^{-1}	1.81×10^{-6}	1.65×10^{-11}
${SO_4}^{2-}$, mol L^{-1}	4.82×10^{-5}	3.19×10^{-3}
CO_2, atm	0.01	0.01
O_2, atm	0.20	0.00
N_2, atm	0.79	0.80
pH	5.0	7.0

[a]The water table is the lower boundary. The water table is a free exit boundary for the aqueous phase transport and a no-flow boundary for gaseous phase transport.

at the lower end of the column. Aqueous flow is included but dispersion in the aqueous phase is neglected to avoid nonphysical results due to the application of macrodispersivities at local-scale reaction zones (Table 3). If calcite is present in the column, acidity produced by reaction (20) is neutralized, leading to the production of carbon dioxide, which partially compensates total pressure reduction by oxygen depletion:

$$CaCO_3 + 2H^+ \leftrightarrow Ca^{2+} + CO_2 + H_2O \tag{22}$$

[43] The simulations include 8 components (Fe^{2+}, H^+, SO_4^{2-}, CO_3^{2-}, Fe^{3+}, $O_2(aq)$, $N_2(aq)$, and Ca^{2+}), 12 secondary aqueous species (including HCO_3^-, and H_2CO_3), 3 gas species (O_2, CO_2, and N_2), pyrite and calcite as primary minerals, and ferrihydrite and gypsum as secondary minerals. Table 3 summarizes the parameters used for this simulation, including initial mineral volume fractions and reactions rates, and Table 5 presents initial and boundary conditions for the 8 components.

[44] Two different cases are considered in order to investigate the role of calcite dissolution and precipitation on gas transport in this environment. In the first case, calcite is included with a volume fraction of 30% to represent mine tailings consisting of a carbonate-rich rock (e.g., skarn). In the second case, a carbonate-depleted rock (e.g., granite) is considered and calcite is therefore not included in the simulation. Results are shown at 50 years.

[45] For both scenarios, simulation results show that as O_2 is consumed at depth, N_2 and CO_2 concentrations increase relative to atmospheric conditions to compensate for depletion of O_2 (Figure 10). However, in the carbonate-rich sediment, the increase in CO_2 is more significant, indicating an increase in the amount of inorganic carbon present in the aqueous and gaseous phases. As a result of acidity production during pyrite oxidation (20), calcite dissolves according to (22) producing CO_2. Additionally, some of the CO_2 produced remains in the aqueous phase, and undergoes aqueous complexation (Figure 11a). The presence of calcite buffers the pH at values around 7, making bicarbonate the most abundant carbonate complex, while its absence results in very low pH values (Figure 11b).

[46] In the carbonate-rich system, advection accounts for 9.5% of the net flux of atmospheric oxygen into the column, while diffusion accounts for the remaining 90.5%, with the nonequimolar component being 3.7% of the net flux (Figure 12a). If the nonequimolar flux component were not considered (i.e., using equations (11) and (12) in lieu of (7) for gas diffusion), the advective flux would be greater to compensate for the nonseparative flux that is accounted for by the nonequimolar component of diffusion in the simulations presented here. Therefore the relative importance of advection is reduced by the existence of the nonequimolar flux.

[47] For the carbonate–depleted system, advection is greater than in the carbonate-rich system and it gains relative importance in the distribution of flux components: 15.8% for advection, 84.2% for diffusion, with the nonequimolar component being 4.2% of the net oxygen flux (Figure 12b). Very similar results are obtained for the carbonate-rich system when $CO_2(g)$ is not included in the simulation (16.4–83.6–3.6%) although calcite is present. This indicates that the $CO_2(g)$ buildup is caused partially by advective transport, but mostly by calcite dissolution. Overall, the results compare well with findings by *Binning et al.* [2007] for a binary system including N_2 and O_2; and confirm that reaction-induced gas advection can significantly contribute to supply oxygen for pyrite oxidation. At quasi steady state, oxygen diffusive fluxes into the column are

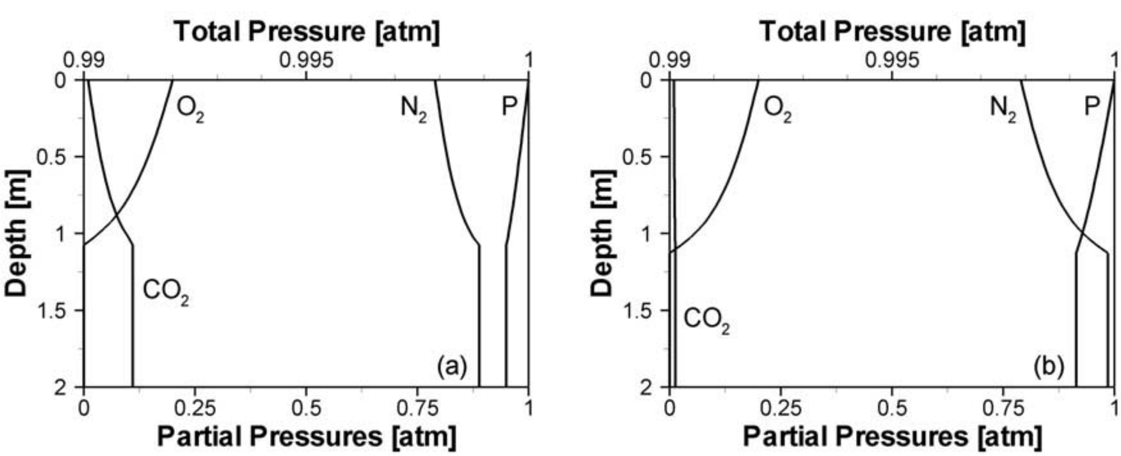

Figure 10. Simulated total and partial pressures during pyrite oxidation in (a) a carbonate-rich system and (b) a carbonate-depleted system.

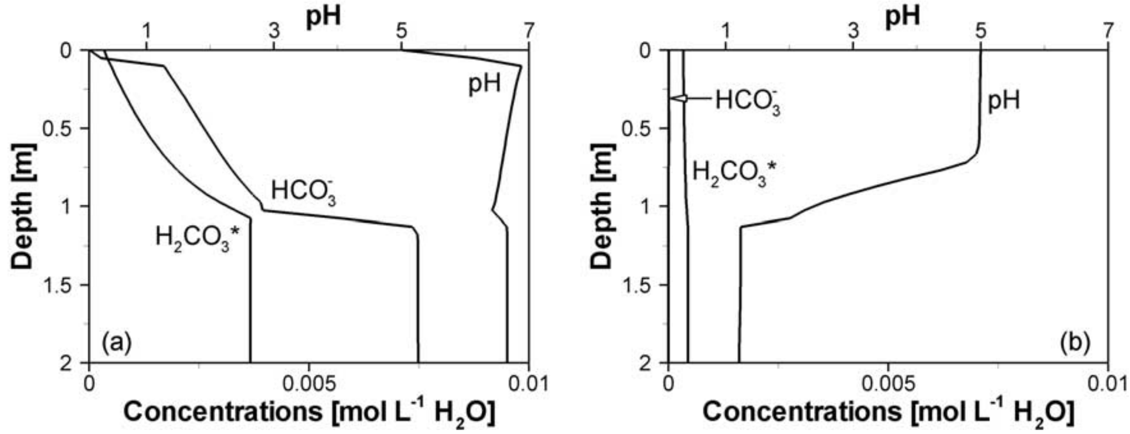

Figure 11. The pH and concentrations of the carbonate aqueous species in (a) a carbonate-rich system and (b) a carbonate-depleted system.

approximately 4 times larger than nonseparative fluxes; the nonequimolar and advective fluxes account for 20% of the net oxygen flux.

[48] Aqueous flow also contributes to the transport of chemical components in the column. For gases with low solubility such as O_2, the contribution of the aqueous flux is small in comparison with the gaseous flux. For example, in the carbonate-rich system, at the top of the column, aqueous influxes represent 0.11% of net gaseous influxes. However, for gases with high solubility such as CO_2, which in the aqueous phase speciates as CO_3^{2-}, HCO_3^-, and H_2CO_3, aqueous contributions can be relevant. In the carbonate-rich system, aqueous advective influx of total inorganic carbon reaches 23% of the advective gas influx at the top of the column. It is, however, overwhelmed by the diffusive outflux of CO_2, which is two orders of magnitude larger than gas ingress in the aqueous phase. In the calcite-depleted system, diffusion and advection in the gas phase approximately balance each other, and aqueous advection is the only mechanism that supplies inorganic carbon into the column at a rate of 1.94×10^{-4} mol d^{-1}. Only aqueous fluxes occur across the water table at the lower end of the column. In the carbonate-rich system, the aqueous flux at the water table contributes 12% to the total outflux of inorganic carbon across both boundaries.

6. Conclusions

[49] This paper has presented the development of a reactive transport model that integrates multicomponent gas diffusion and advection into a multicomponent reactive transport code. The resulting model facilitates the investigation of interactions between geochemical reactions, rapid gas transport, and solute transport in the vadose zone. The use of the direct substitution approach allows relatively large time steps and solutions can be achieved with reasonable computational cost.

[50] It was found that reaction-driven advection can contribute significantly to gas transport. This is in agreement with previous studies of pyrite oxidation [*Binning et al.*, 2007], where depletion of oxygen induces gas flow into the reaction zone. In addition, it was found that methane oxidation in unsaturated porous media contributes to the reduction of methane emissions to the atmosphere not only directly by removing methane but also indirectly by changing the gas flow pattern. A reduction in total gas pressure

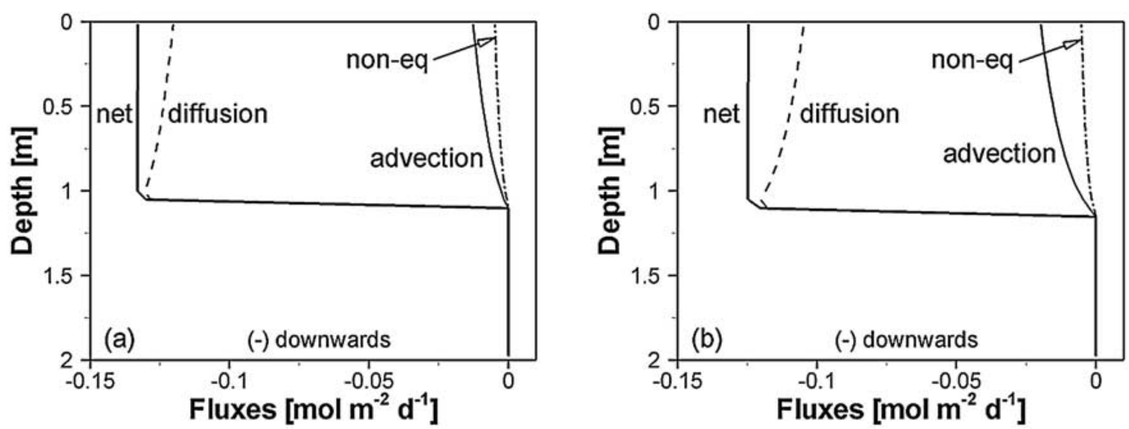

Figure 12. Advective, diffusive, and nonequimolar components of the net flux for O_2 in (a) a carbonate-rich system and (b) a carbonate-depleted system during pyrite oxidation.

caused by methane oxidation may lead to the reduction of the advective gas flow component (landfill cover soils) or even a reversal in the flow direction (e.g., oil spill site).

[51] A model for multicomponent diffusion is required to simulate gas transport adequately and in a comprehensive fashion. Firstly, the assumption, necessary for Fick's law, that diffusing gas species are dilute in the bulk gas phase is not satisfied in any of the case studies. Secondly, reactions change the gas composition and product species have different diffusion properties than reactant species. Therefore, considering a single diffusion coefficient for all gases using Fick's law may lead to errors in the calculation of diffusive fluxes in systems where reactions occur.

[52] Knudsen diffusion does not appear to play a significant role in the systems studied here. Given the range of permeabilites encountered in the examples (10^{-11}–10^{-14} m^2), this result is in agreement with results from other works [*Massmann and Farrier*, 1992; *Sleep*, 1998]. Nevertheless, the use of the dusty gas model equations allowed to account for the nonseparative transport mechanism associated with diffusion: the nonequimolar flux component. This flux component depends on the composition of the gas mixture and is therefore affected by changes brought about by chemical reactions. For example, oxidation of CH_4 with a molecular weight of 16 g mol^{-1} to CO_2 with a molecular weight of 44 g mol^{-1} can change the magnitude or even reverse the direction of the nonequimolar flux component. In systems where sulfide mineral oxidation plays a role, the nonequimolar flux component can also be significant because oxygen consumed by the reaction has a higher molecular weight relative to air.

[53] The inclusion of the aqueous phase, as well as of a number of homogeneous and heterogeneous reactions, allows for the consideration of processes such a pH buffering and aqueous speciation, which were not included in previous gas transport models. Such processes may also affect gas transport. In the case of pyrite oxidation, the presence of calcite accounts for higher CO_2 concentrations, and decreases the contribution of advection to the net oxygen flux into the soil column. Aqueous advection can be a relevant transport mechanism for gases with high solubility such as CO_2, although in general gas fluxes tend to overwhelm aqueous fluxes. On the other hand, at the oil spill site, concentrations of gas species are not significantly affected by calcite dissolution although measured pH values and solution composition cannot be simulated satisfactorily without the inclusion of calcite and carbonate speciation in the model.

Appendix A: Physical Relationships

[54] Gas density is a function of gas concentrations and molecular weights:

$$\rho_g = \sum_{i=1}^{N_g} M_g^i c_g^i \tag{A1}$$

The viscosity of the gas phase (μ_g) is computed using the following semiempirical expression based on the kinetic theory of gases [*Bird et al.*, 2002]:

$$\mu_g = \sum_{i=1}^{n} \frac{\chi_g^i \mu_g^i}{\sum_{j=1}^{n} \chi_g^j \phi_{ij}} \tag{A2}$$

where

$$\phi_{ij} = \frac{1}{\sqrt{8}} \left(1 + \frac{W_g^i}{W_g^j}\right)^{-1/2} \left[1 + \left(\frac{\mu_g^i}{\mu_g^j}\right)^{1/2} \left(\frac{W_g^j}{W_g^i}\right)^{1/4}\right]^2 \tag{A3}$$

Relative permeability in the gas phase (k_{rg}) is calculated using the relationship by *Parker et al.* [1987]:

$$k_{rg} = (1 - S_{ea})^{1/2}\left(1 - S_{ea}^{1/m}\right)^{2m} \tag{A4}$$

with $m = 1 - 1/n$, where n and m are soil hydraulic function parameters. S_{ea} is the effective saturation of the aqueous phase defined by

$$S_{ea} = \frac{S_a - S_{ra}}{1 - S_{ra}} \tag{A5}$$

where S_{ra} is the residual saturation in the aqueous phase (dimensionless).

[55] Tortuosity τ_g (dimensionless) is expressed as a function of aqueous phase saturation and porosity based on the semiempirical expression by *Millington* [1959]:

$$\tau_g = S_g^{7/3} \phi^{1/3} \tag{A6}$$

or by the expression by *Moldrup et al.* [2000] for repacked soils:

$$\tau_g = S_g^{3/2} \phi^{1/2} \tag{A7}$$

The expression found by *Heid et al.* [1950] for dry consolidated materials has been widely used in the literature to estimate the Klinkenberg parameter:

$$b_{ref} = b_{air} = 0.11\left(k_{rg}k\right)^{-0.39} \tag{A8}$$

The correlation recently found by *Reinecke and Sleep* [2002] can be used for nonconsolidated finer grained materials under partially saturated conditions:

$$b_{ref} = b_{air} = 5.57\left(k_{rg}k\right)^{-0.24} \tag{A9}$$

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